

WASEDA University
Graduate School of Information, Production and Systems

Diagnosis-Guided Augmentation and Frequency-Gated Fusion for Actor-Conditioned Human-Value Recognition

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Midterm Presentation · Focus: WAIPS-03

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Whose values does a news report describe?

Chile protests: a reported government statement

> *"He justified the decision, saying he wanted to maintain **public order** against vandals and crime."*

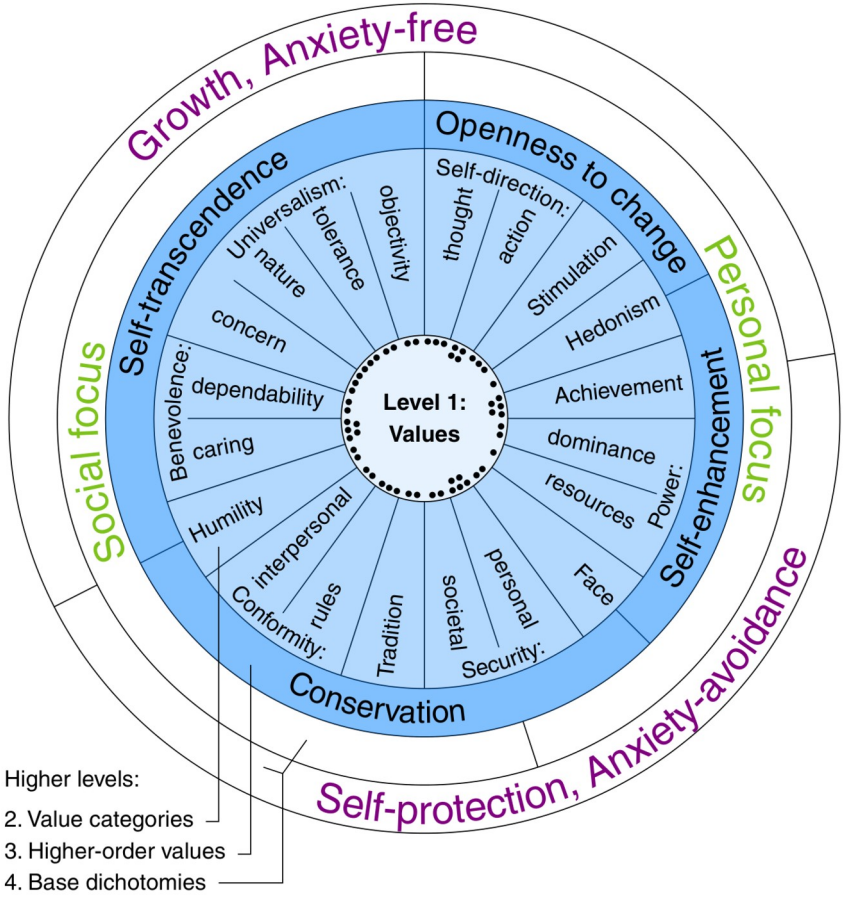
Whose conduct?	Which value?	Which direction?
Government and Security Forces	Have a safe country	**Aligned**

Aligned = supports the value. Contradictory = goes against it.

The benchmark labels reported conduct in its news context.

Source excerpt and reference: Huang, section 5.9, Table 9 (Chile case).

A shared vocabulary for human values



54 specific values

20 broader categories

Values describe what people consider important.

Example: Security, societal

Have a safe country

Have a stable society

Kiesel et al. (ACL 2022),
mainly based on Schwartz's value theory.

Figure: Kiesel et al., ACL 2022, Fig. 1. aclanthology.org/2022.acl-long.306/

NTCIR-19 FEHU: a shared evaluation task

Fact-based Event-centric Human-value Understanding

Values in factual news, at article and event levels

Organizing institutions

University of Tsukuba
Yao Wang, Zhuochen Liu,
Jiankang Chen, Haitao Yu

AIST (Japan)
Xin Liu, Kyoungsook Kim

University of Innsbruck
Adam Jatowt

Shared-task timeline

Date	Milestone
Sep 2025	Task launch
Jan 2026	Training data
Aug 2026	Results returned
8–10 Dec 2026	Conference, Tokyo

Sources: FEHU organizers / important dates; NTCIR-19 conference. Checked 1 Oct 2026.

My task: labels for each supplied actor

Input

News article

Supplied actor list



For each actor

Relevant value + direction

Aligned / contradictory

Several labels may apply

54 values × 2 directions = 108 possible labels

Evidence comes from the article; actor identities are supplied.

Task definition: FEHU official resources; Huang, section 3.1.

What these labels could help readers do

Find reports about a value and an actor

"Find reports about public security and government actions."

Compare how reports portray institutions

Keep each judgment connected to the right person or organization.

A useful label must preserve who did what.

Application motivation: Huang, introduction. Reader question is illustrative.

Research goal: fewer wrong and missed labels

Miss a relevant value

The model may overlook
a value in an actor's conduct.

Add an incorrect label

A plausible interpretation
may disagree with the reference.

Questions for this talk

How should support from different predictions guide revision?

Which revision steps reduce errors?

Goal: recover relevant labels while controlling incorrect predictions.

Huang, sections 1 and 3.3. This talk focuses on the fusion research question.

How the classifier supports the decision

Approach	Role of the trained classifier
EAVIT	Narrow value definitions before the language model decides
WAIPS-03	**Help revise labels after the initial predictions**

Label selection and direction checking use separate decisions.

My design uses classifier scores to support selective revision.

Zhu et al., EAVIT (IJCAI 2025); Huang, section 2. doi.org/10.24963/ijcai.2025/934

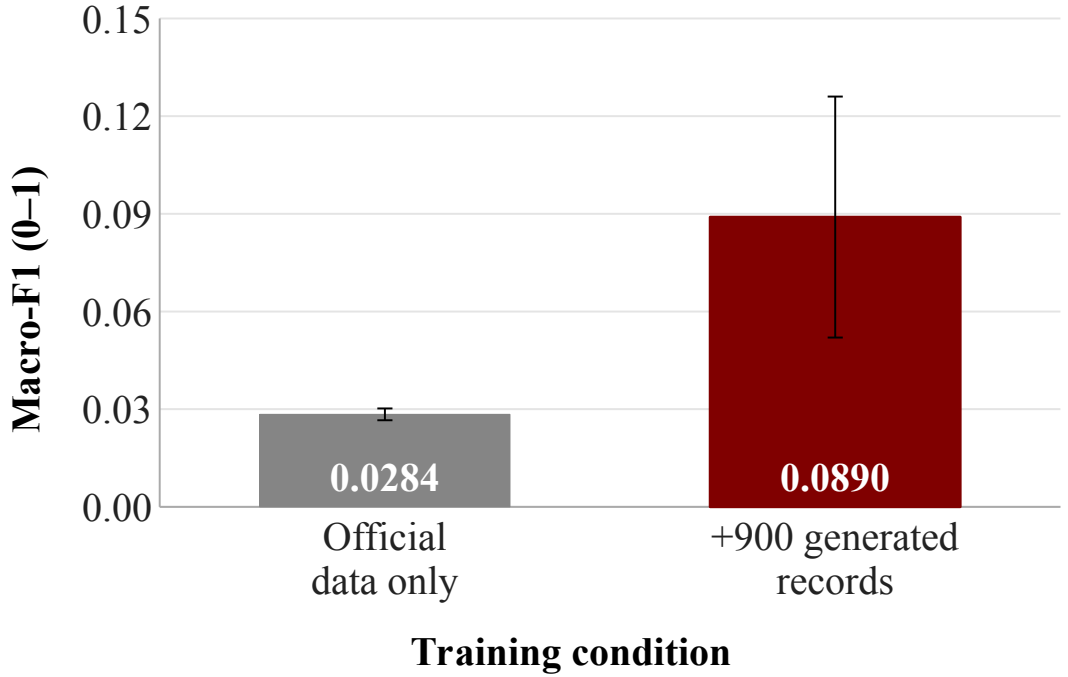
Data augmentation: method and initial evidence

Target 18 missed directed labels

Generate 900 news-like examples linking actors to values and directions.

Train the same DeBERTa classifier under both conditions, with 3 seeds.

Target F1 remains 0.0319 (threshold 0.30).
DRR worsens: 0.1580 to 0.2823.



Three-run means. Error bars: sample SD.

Macro-F1 improves, while the targeted labels remain difficult.

Huang, Table 5, A vs B. Independent training study, distinct from the historical export on slide 15.

WAIPS-02: a simpler comparison system

How it selects labels

Keep a directed label supported by at least 2 of 3 prompts.

Use a separate prediction to choose its direction.

What the comparison tests

Does selective fusion improve recognition

beyond this majority-based approach?

WAIPS-02 provides a simpler reference for the WAIPS-03 results.

Huang, section 4.4: three Pro prompt variants, followed by a Flash direction lookup.

WAIPS-03 combines scores and prompt predictions

PROPOSED METHOD · ACTUAL INPUTS AND THEIR ROLES

Input	Role in fusion
Initial prediction	Propose labels from generated event descriptions
Augmented classifier	Score candidates using learned label patterns
Four auxiliary predictions	Supply candidates and agreement, including Liu's targeted prompt

WAIPS-02 informs the design: select values first, then check direction.

Its final output is not an input to WAIPS-03.

Fusion decides which initial labels to revise using these supporting signals.

Huang, sections 1.3 and 4.5–4.7. WAIPS-03 = frequency-gated fusion, E7-G4.

Support scores combine votes and classifier estimates

For one actor and a candidate value–direction label (v, d)

Removal score $s_{rm} = e + 0.25\ o + \max(p_0, p_1)$

Addition score $s_{add} = e + p_d$

e: auxiliary votes for the exact value and direction

o: votes for the same value with only the opposite direction

p_0, p_1: classifier estimates for the two directions. Four auxiliary predictions vote.

Removal accepts partial value support. Addition uses exact direction support.

Huang, §4.6, support-score equations. Actor and value indices omitted for readability.

Calibrated decisions and direction checks

$$q_{rm} = C_{rm}(s_{rm}) \qquad q_{add} = C_{add}(s_{add})$$

C: calibration from development labels. q: estimated correctness.

Remove when $q_{rm} < \theta$

Preserve at least 1 initial label.

Add when $q_{add} \geq \theta$

Absent value, at most 2 labels.

Eligibility: $n_\ell > 130$ training positives. $\theta = F1_{fit}(\text{initial}) / 2$.

Direction QA: one current direction + opposite auxiliary support.

Frequency and count limits constrain revision. The threshold is a heuristic.

Huang, §§4.6–4.7 and Algorithm 1. n is label frequency. Full implementation: backup 24.

Evaluation: recognition and direction errors

Articles: 1,228 training 177 development 175 official test

$$\text{Micro-F1} = 2TP / (2TP + FP + FN)$$

TP: correct, FP: extra,
FN: missed labels

$$\text{Macro-F1} = (\sum_{\ell \in L} F1_{\ell}) / |L|$$

L: labels with gold examples
Each label has equal weight.

$$DRR = R / |E|$$

E: pairs with one gold direction
R: number reversed within E

Higher F1 means better recognition. Lower DRR means fewer reversals.

Huang, §5.2, Micro-F1, gold-supported Macro-F1 and direction reversal rate (DRR).

Recognition scores across four approaches

EVALUATION · TASK 1b, WITH A REPORTED PURE-PROMPT REFERENCE

Method	Micro-F1 ↑	Macro-F1 ↑	DRR ↓
Augmented classifier (historical)	0.3306	0.1303	0.2880
Pure prompt*	0.3787	0.2754	—
WAIPS-02	0.5560	0.3816	0.0338
WAIPS-03: proposed fusion	**0.5817**	**0.4308**	**0.0419**

* Pure prompt: author-reported external score. — = DRR not provided.

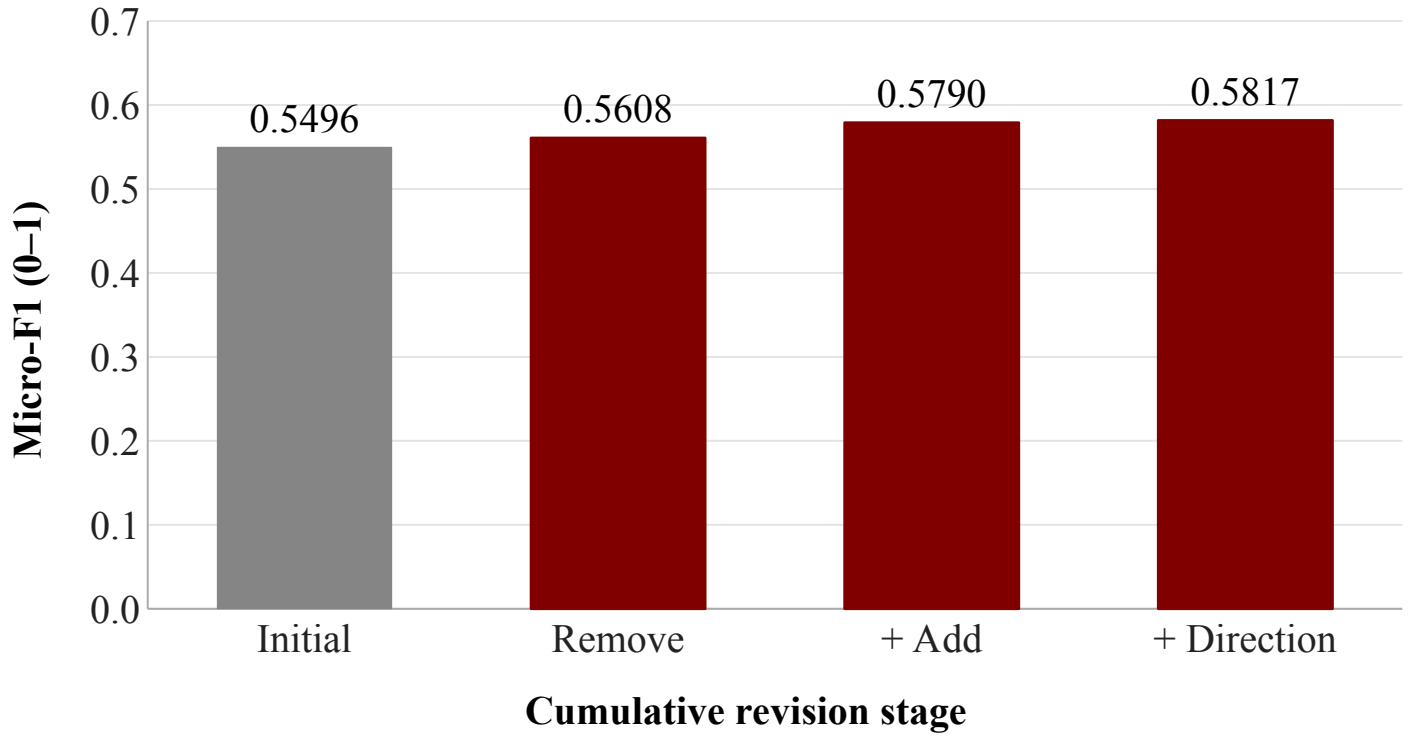
Other rows: 177 dev articles. Encoder: 456 actors, WAIPS: 459. Scores: 0–1.

Fusion improves F1 over WAIPS-02, with a higher reversal rate.

Huang, Table 4 and archived metrics. Pure-prompt F1 provided by Peiyang Huang on 1 Oct 2026.

Which revisions bring the improvement?

EVALUATION · FIXED INITIAL OUTPUTS ON 177 DEVELOPMENT ARTICLES



−170

unmatched labels

−50

missed labels

Net changes from initial

Removal and addition provide most of the gain. Direction checking adds little.

Huang, Table 7, G0/G1/G3/G4. Net label counts from the same saved predictions and development gold.

Official test: better F1, remaining direction errors

EVALUATION · SEPARATE TEST SET, 175 ARTICLES

Submitted system	Micro-F1 ↑	Macro-F1 ↑
WAIPS-02: majority and direction choice	0.5496	0.3580
WAIPS-03: proposed fusion	**0.5808**	**0.3952**

Direction reversal rate rises: 0.0395 to 0.0477.

More predictions oppose the reference direction.

Overall recognition improves, while direction reliability remains a priority.

Official FEHU overview, Table 6; Huang, Table 8. Scores 0–1; lower reversal rate is better.

A direction check can introduce an error

Actor: Government and Security Forces · Value: Have a safe country

> *"...he wanted to maintain **public order** against vandals and crime."*

Before checking	After checking	Reference
Aligned	Contradictory	Aligned

The saved answer contains the direction, without an explanation.

Direction revision can overturn a correct initial judgment.

Huang, section 5.9, Table 9: Chile, 0-4716-3-1, actor -1, value 26. Excerpt shortened.

Next: evidence for more reliable direction changes

FUTURE WORK · PLANNED VALIDATION

A proposed change should cite the actor's conduct

Compare useful corrections and introduced errors
with their supporting passages.

Test on separate evaluation articles

Measure harmful revisions and missed corrections,
alongside overall label recognition.

The aim is fewer harmful changes while keeping useful corrections.

Planned work, extending Huang, sections 6.1–6.2. Effectiveness is not yet established.

Beyond FEHU: value analysis across news sources

FUTURE WORK · PLANNED TRANSFER AND APPLICATION

Research step	Question to test
New sources and events	Does revision still help on news outside the FEHU collection?
Researcher review	Do labels and supporting passages help compare actors' conduct?
Human evaluation	Are labels and passages correct? Does review take less time?

Test transfer to new material and usefulness for human analysis.

Proposed research extension based on Huang, §6.2 evidence evaluation; no transfer result yet.

Conclusion: selective fusion improves recognition

Research process

Training augmentation and prompt aggregation inform WAIPS-03.

Fusion combines classifier scores with auxiliary predictions.

Main result

Higher development and official-test F1 than WAIPS-02

Fewer unmatched and missed labels than the initial prediction

Next: stronger text evidence and evaluation beyond FEHU.

Huang, Tables 4 and 7–9. Main talk ends here.